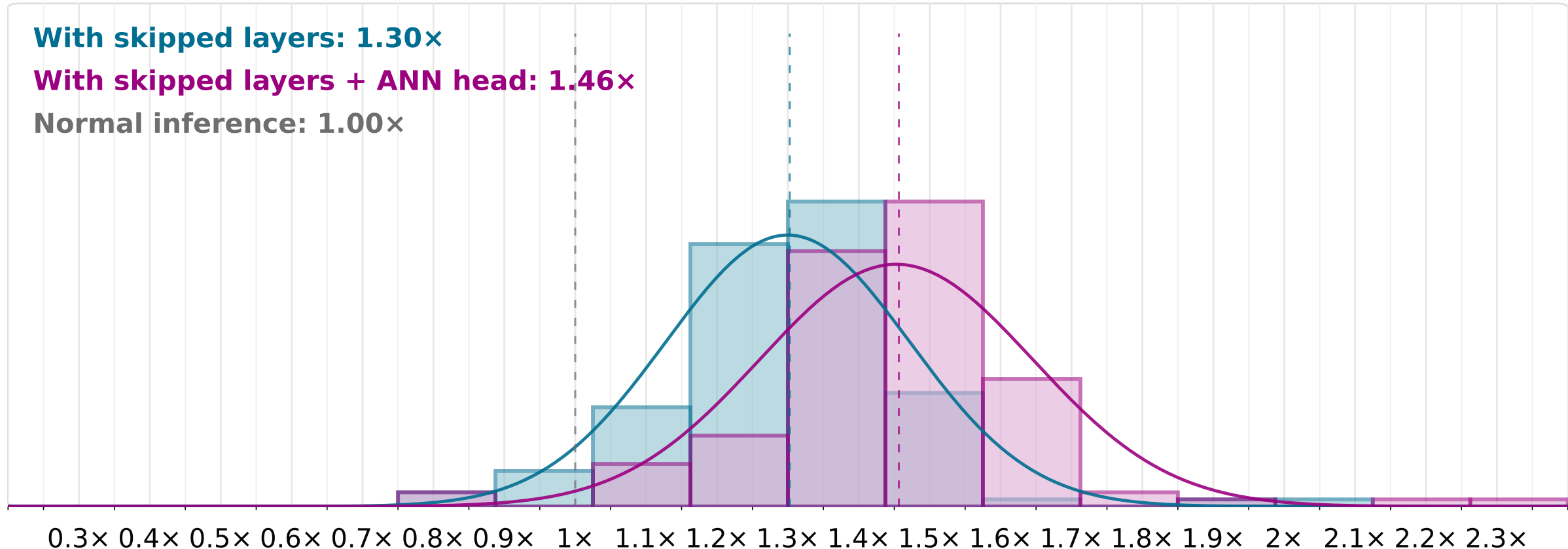


Self-speculation per-token speedup

Llama 3.2 3B Instruct



Setup

Prompt set = **concrete**
Gap = **(1, 1)**
Block size = **2**
Layers = **28**
LM vocab = **128,256**
LM params = **3.21B**
Head params = **394.0M**
Head portion = **12.26%**

Runtime

GPU = **NVIDIA L4**
Backend = **sdpa**
Model dtype = **bfloat16**
Bridge dtype = **bfloat16**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **120**

Results

Peak mem normal = **6.05 GiB**
Mean speedup (w/o ANNH) = **1.30×**
Acceptance rate (w/o ANNH) = **44.2%**
Exact match (w/o ANNH) = **49.2%**
Peak mem self (w/o ANNH) = **6.06 GiB**
Mean speedup (w ANNH) = **1.46×**
Acceptance rate (w ANNH) = **43.9%**
Exact match (w ANNH) = **49.2%**
Peak mem self (w ANNH) = **6.11 GiB**
Tokens gen = **s 20,116 / n 20,231**

Profile

Drafter split (w/o ANNH) = **B 41.4% / H 52.7% / O 5.9%**
Verifier/normal (w/o ANNH) = **1.05×**
Drafter/normal (w/o ANNH) = **19.9%**
Drafter split (w ANNH) = **B 75.6% / H 13.4% / O 11.0%**
Verifier/normal (w ANNH) = **1.06×**
Drafter/normal (w ANNH) = **11.5%**
ANNH accuracy = **99.4%**
ANNH head speedup = **6.76×**
ANNH index = **8016 clusters; top-k 100**